



Breathe, Screen, Live

Understanding & Managing Diabetes

A practical guide to living well with diabetes — from diagnosis to daily management

- What diabetes is and the different types
- How to monitor your blood sugar effectively
- Diet, exercise, and medication management
- Warning signs, complications, and when to see your doctor
- South African context: access to care and support

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What Is Diabetes?

Diabetes is a chronic condition that affects how your body turns food into energy. When you eat, your body breaks down carbohydrates into glucose (sugar), which enters your bloodstream. Your pancreas produces insulin — a hormone that acts like a key, helping glucose enter your cells to be used for energy.

In diabetes, either your pancreas doesn't produce enough insulin, or your body's cells don't respond to insulin properly. This causes glucose to build up in your bloodstream, leading to high blood sugar levels.

There are three main types of diabetes:

Type 1 Diabetes

Type 1 diabetes is an autoimmune condition where the immune system attacks and destroys the insulin-producing cells in the pancreas. It typically develops in children and young adults, though it can appear at any age.

People with Type 1 diabetes require daily insulin injections or an insulin pump to survive. There is currently no cure, but with proper management, people with Type 1 diabetes can live long, healthy lives.

Symptoms often appear suddenly and include excessive thirst, frequent urination, unexplained weight loss, and extreme fatigue.

Type 2 Diabetes

Type 2 diabetes is the most common form, affecting about 90% of people with diabetes. It occurs when your body becomes resistant to insulin or doesn't produce enough insulin. It's strongly linked to lifestyle factors including diet, physical activity, and weight.

Type 2 diabetes often develops gradually over many years. Early symptoms may be mild or go unnoticed, which is why regular screening is important — especially if you have risk factors such as being over 45, having a family history of diabetes, being overweight, or having high blood pressure.

Management includes lifestyle changes, oral medications, and sometimes insulin. Many people with Type 2 diabetes can achieve good control through diet and exercise alone, especially when diagnosed early.

Gestational Diabetes

Gestational diabetes develops during pregnancy and usually resolves after childbirth. However, it increases the risk of developing Type 2 diabetes later in life for both mother and child.

Screening for gestational diabetes is typically done between 24 and 28 weeks of pregnancy. Management focuses on blood sugar control through diet, exercise, and sometimes medication to protect both mother and baby.

Monitoring Your Blood Sugar

Regular blood sugar monitoring is the cornerstone of diabetes management. Your healthcare provider will recommend a target range for your blood sugar levels. Generally, fasting levels should be between 4.0 and 7.0 mmol/L, and post-meal levels should be below 10.0 mmol/L.

Keep a log of your readings — note the date, time, reading, and any factors that may have affected it (what you ate, exercise, stress, illness). This information helps you and your doctor make informed decisions about your treatment plan.

Your doctor may also recommend a HbA1c test every 3 to 6 months. This measures your average blood sugar control over the past 2-3 months. The general target is below 7% (53 mmol/mol).

Healthy Eating for Diabetes

Eating well with diabetes doesn't mean restrictive diets. Focus on:

- Choose whole grains over refined carbohydrates (brown rice, oats, wholewheat bread)
- Eat plenty of vegetables — at least 5 portions a day
- Include lean protein with every meal (chicken, fish, eggs, beans, lentils)
- Limit sugary drinks, sweets, and processed foods
- Watch portion sizes — even healthy foods can raise blood sugar in large amounts
- Stay hydrated with water or unsweetened beverages
- Consider the glycaemic index (GI) of foods — low GI foods release sugar more slowly

Physical Activity

Regular physical activity helps your body use insulin more effectively and lowers blood sugar levels. Aim for at least 150 minutes of moderate exercise per week — that's about 30 minutes, 5 days a week.

Good options include brisk walking, swimming, cycling, dancing, or gardening. Start slowly if you've been inactive and gradually increase intensity. Always check with your doctor before starting a new exercise program.

Check your blood sugar before and after exercise, especially if you take insulin or certain diabetes medications, as exercise can cause hypoglycaemia (low blood sugar).

Medication Management

If lifestyle changes aren't enough to control your blood sugar, your doctor may prescribe medication. Common options include:

- Metformin — the most common first-line medication for Type 2 diabetes
- Sulphonylureas — help your pancreas produce more insulin
- DPP-4 inhibitors — help lower blood sugar without causing hypoglycaemia
- SGLT2 inhibitors — help your kidneys remove excess glucose through urine
- Insulin therapy — essential for Type 1 diabetes and sometimes needed for Type 2

Warning Signs & Complications

Untreated or poorly managed diabetes can lead to serious complications. Watch for:

- Frequent infections or slow-healing wounds
- Numbness or tingling in your hands or feet (neuropathy)
- Vision changes or blurred vision (retinopathy)
- Kidney problems (nephropathy)
- Heart disease and stroke — diabetes significantly increases your risk

Regular check-ups with your healthcare team can catch complications early when they're most treatable. This includes annual eye exams, foot checks, kidney function tests, and cardiovascular screening.

In the South African Context

South Africa has the highest rate of diabetes in Africa, with an estimated 4.5 million adults living with the condition. Access to care varies widely, but support is available:

- **Public healthcare:** Diabetes care is available at primary healthcare clinics nationwide. Medications on the Essential Medicines List are provided free of charge at public facilities.
- **Private healthcare:** Medical aid schemes typically cover diabetes management, including medication, glucose monitoring supplies, and annual screening.
- **Support organisations:** Diabetes South Africa (www.diabetessa.org.za) offers education, support groups, and advocacy for people living with diabetes.
- **Community health workers:** Many communities have trained health workers who can provide education and support for diabetes management.

When to See Your Doctor

Contact your healthcare provider immediately if you experience:

- Blood sugar consistently above 15 mmol/L or below 4 mmol/L
- Symptoms of diabetic ketoacidosis (DKA): nausea, vomiting, abdominal pain, fruity-smelling breath, rapid breathing
- Signs of infection: fever, redness, swelling, or pus around a wound
- Vision changes, chest pain, or shortness of breath
- Numbness or tingling that interferes with daily activities

Disclaimer

This guide is for informational purposes only and does not constitute medical advice. Always consult a qualified healthcare professional for diagnosis, treatment, and management of any health condition. Aliento Health and Dr. Leegale Adonis accept no liability for actions taken based on the information in this guide.

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